

SALES FORECASTING USING MACHINE LEARNING AND POWER BI

PROJECT REPORT

Submitted by

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CERTIFICATE

This is to certify that the project report entitled “Sales Forecasting Using Machine Learning and Power BI” is a bona fide work carried out by Chinmayi K, a student of III BCA, Department of Computer Science, JSS College for Women, Saraswathipuram, Mysore, during the academic year 2026–2027. This report is submitted as part of the academic requirements of the BCA programme.

Student Signature: _____

Department Seal: _____

DECLARATION

I, Chinmayi K, hereby declare that the project entitled “Sales Forecasting Using Machine Learning and Power BI” is my original academic work. The work presented in this report has not been submitted elsewhere for any other degree, diploma, or academic award. The sources of information used in the project have been acknowledged in the references section.

Place: Mysore

Date: _____

Signature: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to Jyesta Corporate Entity for providing me with the opportunity to work on the Sales Forecasting project and for supporting my learning throughout its development. I also thank JSS College for Women, Saraswathipuram, Mysore, for providing academic encouragement and a supportive learning environment. I am grateful to my classmates, family members, and everyone who encouraged and supported me during the completion of this project.

ABSTRACT

Sales forecasting is the process of estimating future sales by studying historical sales information and relevant business attributes. Accurate forecasts help organisations plan inventory, allocate resources, set targets, and make informed decisions. This project develops a sales forecasting system using Python, machine learning, Flask, and Power BI.

The sales dataset is cleaned and analysed using Python. Important attributes such as order date, product information, category, sub-category, and region are used as input features. A machine-learning pipeline is trained and saved as a reusable model. The model is tested through a prediction program and integrated with a Flask-based web application. Power BI is used to create interactive visuals for total sales, sales by category, sales by region, sales trends, and related business insights.

The project demonstrates an end-to-end data science workflow, from data preparation and analysis to model development, prediction, visualisation, and presentation.

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1. INTRODUCTION

Businesses generate large volumes of sales data through customer orders, products, locations, and time-based transactions. Converting this information into useful predictions is important for planning and decision-making. Sales forecasting estimates future sales using historical patterns and explanatory variables.

This project combines machine learning and business intelligence. Python is used for data processing, model training, and prediction. Flask provides a simple web interface through which a user can enter sales-related details and receive a forecast. Power BI presents the dataset through interactive charts and summary cards.

The project is designed as an educational implementation of an end-to-end sales analytics and forecasting workflow.

2. OBJECTIVES

To collect and prepare a sales dataset for analysis.

To perform exploratory data analysis and understand sales patterns.

To build a machine-learning model for predicting sales.

To save the trained model and reuse it for new predictions.

To develop a simple Flask web application for user input and prediction.

To create an interactive Power BI dashboard for business insights.

To organise all project files in a clear and reusable structure.

3. PROBLEM STATEMENT

Manual analysis of sales records is time-consuming and may not reveal useful patterns quickly. Businesses need a systematic method to analyse sales data, estimate expected sales, and present important results in an understandable form.

The problem addressed by this project is the development of a simple system that accepts relevant sales information, generates a model-based sales estimate, and presents historical sales information through an interactive dashboard.

4. SCOPE OF THE PROJECT

The project focuses on a structured sales dataset containing fields such as order date, category, sub-category, product information, region, and sales. It includes data preparation, exploratory analysis, model training, prediction, a web interface, and Power BI visualisation.

The system is suitable for academic demonstration and small-scale analytical use. Forecast quality depends on the amount, quality, and representativeness of the available data.

5. LITERATURE REVIEW

Sales forecasting is commonly performed using statistical methods, time-series techniques, and machine-learning algorithms. Traditional methods focus mainly on historical time patterns, while machine-learning methods can use multiple categorical and numerical features.

Data preprocessing is important because missing values, inconsistent formats, and categorical variables can affect model performance. Encoding methods allow categorical information to be used by machine-learning algorithms. Visual analytics tools support decision-making by presenting trends and comparisons clearly.

This project combines these ideas in a practical workflow using Python, scikit-learn, Flask, and Power BI.

6. SYSTEM REQUIREMENTS

Hardware: A computer or laptop with at least 4 GB RAM, sufficient storage, and a modern processor.

Software: Windows operating system, Python 3.x, PyCharm or another Python IDE, Power BI Desktop, and a web browser.

Python libraries: pandas, numpy, scikit-learn, joblib, matplotlib, and Flask.

Input: A CSV sales dataset. Output: Predicted sales values, a web-based prediction result, and Power BI dashboard visuals.

7. DATASET DESCRIPTION

The dataset is stored in CSV format and contains sales-related records. The available fields include Category, City, Country, Order Date, Product Name, Region, Sales, Segment, and Sub-Category. The exact records and values depend on the dataset used in the project.

Order Date represents the date associated with the transaction. Category and Sub-Category describe the product grouping. Product Name identifies the product. Region, City, and Country provide location information. Segment describes the customer or market segment. Sales is the target value used for analysis and model development.

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8. METHODOLOGY

Step 1: Import the CSV dataset using pandas.

Step 2: Inspect column names, data types, missing values, and summary statistics.

Step 3: Clean the data and prepare the features required for modelling.

Step 4: Separate input features from the target sales column.

Step 5: Encode categorical variables using a preprocessing pipeline.

Step 6: Split the data into training and testing sets.

Step 7: Train a regression model and evaluate its predictions.

Step 8: Save the trained model as sales_model.pkl using joblib.

Step 9: Load the model in predict.py and test it with new input data.

Step 10: Integrate the model with a Flask web application.

Step 11: Import the dataset into Power BI and create dashboard visuals.

Step 12: Save screenshots and organise the final project folder.

9. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is used to understand the structure and characteristics of the dataset before modelling. The analysis includes viewing the first rows, checking data types, identifying missing values, and generating descriptive statistics.

Sales can be compared across categories, sub-categories, regions, products, and dates. A time-based chart helps identify changes in sales over time. Category and region charts help identify major contributors to overall sales.

EDA supports better feature selection and helps identify data-quality issues before model training.

10. MODEL DEVELOPMENT

The model is trained using selected input columns and the Sales column as the target. Categorical variables are transformed through encoding so that the model can process them. A scikit-learn pipeline combines preprocessing and the regression model in one reusable object.

The trained pipeline is saved as `model/sales_model.pkl`. Saving the complete pipeline ensures that the same preprocessing steps are applied during prediction.

Model performance can be evaluated using regression measures such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). The exact values depend on the dataset and train-test split.

11. PREDICTION SYSTEM

The predict.py program loads the saved model using joblib. A new pandas DataFrame is created with the same feature names expected by the model. The model then produces an estimated sales value.

The input columns must match the columns used during training. If a required field such as Category or Sub-Category is missing, the pipeline reports an error. Matching the feature names and preprocessing steps ensures successful prediction.

The prediction output is displayed in the Python console and may also be saved to a CSV file for later use.

Sales Forecasting System

Order Date:
08/01/2027

Product:
knofel 25

Category:
Technology

Sub-Category:
phones

Region:
South

Predict Sales

Forecasted Sales: 391.1

12. FLASK WEB APPLICATION

The Flask application is implemented in main.py. The application loads the saved machine-learning model and provides a home route that displays index.html from the templates folder.

The web form accepts relevant inputs such as order date, product information, category, sub-category, and region. When the user clicks the prediction button, Flask collects the form values, creates a DataFrame, calls the model, and displays the forecasted sales result.

The static folder contains style.css, which improves the appearance of the webpage. The application can be accessed locally through the Flask development server.

Sales Forecasting System

Order Date:

Product:

Category:

Sub-Category:

Region:

Predict Sales

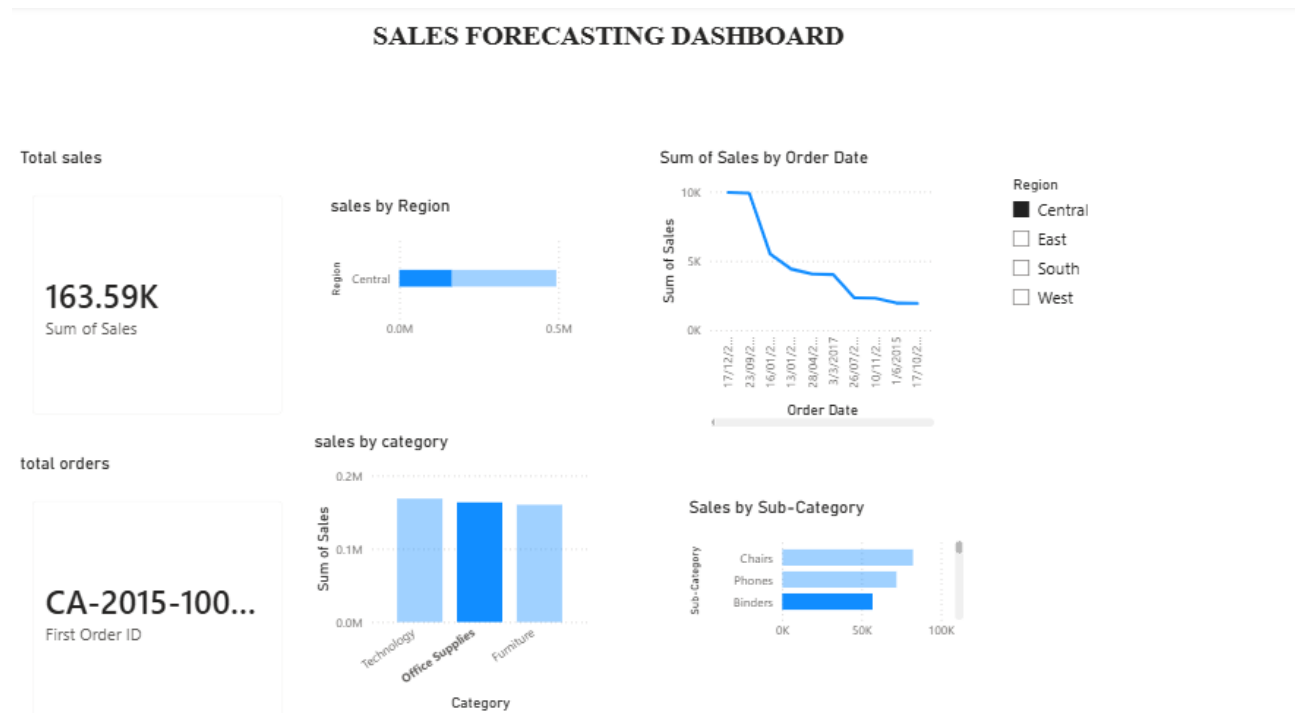
Forecasted Sales: 391.1

13. POWER BI DASHBOARD

Power BI is used to create an interactive dashboard from the sales CSV file. The dashboard provides a visual summary of the dataset and supports quick exploration.

Recommended visuals include a Total Sales card, Sales by Category chart, Sales by Region chart, Sales Trend line chart, Sales by Sub-Category chart, Total Orders card when an order identifier is available, and a Region slicer.

The dashboard is saved as sales_forecasting_dashboard.pbix. The report can be updated when the underlying dataset changes.



14. RESULTS AND DISCUSSION

The project successfully demonstrates a complete workflow for sales analysis and prediction. The trained model is saved and loaded successfully, and the prediction program produces an estimated sales value for valid input data.

The Flask application provides a simple interface for entering input values and displaying the result. The Power BI dashboard presents important sales comparisons and trends in a visual format.

The results show that integrating machine learning with a web interface and business intelligence dashboard can make analytical outputs easier to use. The accuracy of the forecast depends on data quality, feature selection, model choice, and the amount of historical information.

15. CONCLUSION

The Sales Forecasting Using Machine Learning and Power BI project integrates data science, web development, and business intelligence in one application. Python supports data preparation and model development, the saved model supports repeatable predictions, Flask provides an accessible user interface, and Power BI provides interactive visual analysis.

The project meets its main objectives of analysing sales data, developing a prediction model, generating forecasted values, and presenting sales information through a dashboard. It also provides a foundation for future improvements using larger datasets and advanced forecasting methods.

16. FUTURE SCOPE

Use a larger and more recent dataset.

Add inventory, discounts, promotions, holidays, and customer information as features.

Compare multiple regression and time-series models.

Add automatic model evaluation and performance monitoring.

Connect the application to a database.

Deploy the Flask application to a cloud platform.

Enable scheduled dashboard refresh and interactive forecasting.

Add user authentication and role-based access for a production system.

17. REFERENCES

Python Software Foundation. Python Documentation.

pandas Documentation.

scikit-learn Documentation.

Flask Documentation.

Microsoft Power BI Documentation.

Joblib Documentation.

18. APPENDIX

Project folder structure:

Sales_Forecasting/

- |— dataset/ (sales CSV file)
- |— model/ (sales_model.pkl)
- |— templates/ (index.html)
- |— static/ (style.css)
- |— screenshots/
- |— train_model.py
- |— predict.py
- |— main.py
- |— eda.py
- |— forecast_result.csv
- |— sales_forecasting_dashboard.pbix

Add the following screenshots before final submission: dataset preview, EDA output, model training output, prediction output, Flask webpage, forecast result, and Power BI dashboard.